# [Candidate Name]

[Location] | [Phone] | [Email] | [LinkedIn] | [GitHub] | [Portfolio] | **Available Now**

## Education

**State Islamic University of Imam Bonjol Padang** August 2022 – September 2026
*Bachelor of Information Systems (Graduated September 2026)* *Padang, Indonesia*

## Research & Academic Engineering

**Empirical Performance & Scaling Analysis of Event-Driven Systems** Undergraduate Thesis
*Undergraduate Thesis Research | Advisor-Approved Study | Go, RabbitMQ, k6, Prometheus* *Padang, Indonesia*

- Built a reproducible Go/RabbitMQ benchmark harness evaluating **34,951 requests at 200 VUs**; cut **P95 latency 58.9%** (135.5ms to 55.7ms) via autoscaling and queue tuning, reaching **96.56 RPS (0% errors)**.
- Conducted fault-injection experiments simulating broker crashes and network partitions; proved zero data loss across **14,170 transactional events** using dead-letter queue routing.

## Experience

**Software Engineer Intern** September 2025 – December 2025
*TIME Language Centre* *Padang, Indonesia (Hybrid)*

- Designed and shipped a 4-service Go/Node.js course platform with **RabbitMQ** domain decoupling, DLQ recovery, **Redis** caching, **Xendit** payments, and **Bunny.net** video; validated via **k6 load tests (0 failed requests)**.
- Deployed services on Kubernetes (K3s) behind an Nginx TLS reverse proxy; configured Prometheus, Grafana, and Loki dashboards for health, latency, and error observability.
- Authored **119+ unit, integration, and E2E tests** across 5 critical workflows; diagnosed single-replica bottlenecks using **k6** load testing prior to horizontally scaled deployment.
- Built an MVC ERP backend that automated course scheduling for **20+ staff**, replacing spreadsheet-based operations and introducing review-gated changes for the service codebase.

**Software Developer (Part-time)** April 2025 – December 2025
*CodeAtHome* *Remote*

- Delivered responsive web features and REST API integrations across **4+ client applications**, consolidating shared data-fetching and UI patterns to accelerate implementation across projects.
- Optimized client-server data synchronization and caching via **TanStack Query**, reducing redundant API calls.
- Crafted a modular, reusable UI component library, reducing boilerplate across client web projects.

**Software Engineer Intern** January 2025 – March 2025
*Auto2000 (Astra International)* *Labuhanbatu, Indonesia (Onsite)*

- Shipped a Next.js dealership platform with an interactive vehicle recommendation flow, integrating vehicle attributes and user preferences into a guided search experience.
- Engineered an automated document expiry reminder system with **RBAC**, eliminating manual tracking.
- Automated cleaning and migration of **15 years** of legacy service records (**38k+/year**) via Python.

## Projects

Additional projects at kiril.is-a.dev/projects

**Slither** | *Go, Docker, Redis, WebSocket, React* GitHub

- Implemented a Go/WebSocket game server using a **2D spatial-hash grid** for collision broad-phase detection; sustained **60 FPS with 10,000+ entities** in a Dockerized test, documented via `pprof` **traces**.
- Eliminated perceived input lag via **client-side prediction** and **input buffering with server reconciliation**; reduced network payload by **78%** using binary **delta compression**.
- Stress-tested containerized browser client under **1,000 concurrent WebSocket connections**; used `pprof`-guided `sync.Pool` buffers and binary serialization to remove observed GC pauses.

**Strata: Fault-Tolerant Distributed Search Engine** | *Go, gRPC, Raft, LSM-Tree* GitHub | Consistency Report

- Built a hand-rolled Go LSM engine (SkipList memtable, SSTable compaction, **Bloom filters**, **Roaring Bitmaps**) with an inverted index layer; replicated across 3 nodes via **Raft consensus**.
- Verified linearizable consistency under network partitions using a **Porcupine** checker; documented benchmark results and failure-case reproducibility in a public **Consistency Report**.

## Open Source Contributions

- **Google Go Client**: Corrected Enterprise billing API types from `int` to `float64`, restoring JSON unmarshalling for `PremiumRequestUsageItem`; merged upstream with regression tests. (Merged PR)
- **OpenTelemetry Collector Contrib**: Merged an upstream change adding exporter queue batching and backpressure control; documented architecture and benchmarks at 50k spans/sec. (Merged PR)
- **Meilisearch Go Client**: Refactored `FacetSearchRequest.Filter` to accept slice inputs, unlocking multi-condition facet filtering; merged upstream with unit test coverage. (Merged PR)

## Honors & Awards

- **AstraPay Hackathon 2026** – Finalist (Top 10 nationally): Built real-time QRIS credit scoring for MSMEs.
- **PIDI DIGDAYA Hackathon 2026** – Semifinalist (Top 400/2,500+): Sandbox work-trials preventing fraud.

## Technical Skills

**Languages**: Go, TypeScript, Python, SQL, Bash | **Cloud & Tools**: Docker, Kubernetes, AWS, GCP, Prometheus, k6
**Technologies**: Redis, RabbitMQ, PostgreSQL, Raft, LSM-Tree, OpenTelemetry, gRPC, WebSockets